Heuristic Evaluation Report

Project: Online Flashcards
URL: https://github.com/alekkiq/online-flashcards
Evaluators: Teemu Laasio, Blendi Grajqevci, Alabbas Alkhrsany, Aleksi Putkonen
Date: 27.04.2026

Overview

The team conducted a comprehensive evaluation of the online-flashcards application on 27.04.2026. The team followed Nielsen's Heuristics as the criteria for finding usability issues in the application. The section and table below represents the usability issues found and their suggested improvements. Each team member did a heuristic evaluation on their own and the table below combines the results of those evaluations.

Heuristic Report (27.04.2026)

The table below represents the usability issue findings. The table includes 5 columns:

- **Heuristic**: Nielsen's heuristic guidelines.
- **Description of the Issue:** A brief description of the problem encountered during testing.
- **Screenshot:** A screenshot of the problem if applicable.
- **Severity:** The severity of the issue, see the ratings below.
- **Suggested Improvement:** The suggested improvements on how to tackle the issue.

Severity ratings:
0 = Not a usability problem
1 = Cosmetic problem
2 = Minor usability problem
3 = Major usability problem
4 = Usability catastrophe

No	Heuristic	Description of the Issue	Screenshot (if applicable)	Severity (0-4)	Suggested Improvement
1	**H1-1: Simple & natural dialog**	A lot of lorem ipsum text found in the project, and some unnecessary links.		1	Fill out with real business text.
		When searching for quizzes and there are no results, there isn't any text saying "No search results found."		1	Change to "No search results found" for the query.
2	**H1-2: Speak the users' language**	Found a piece of untranslated text in the app. At the Flashcard game stage.		3	Localize the text also.
		Forgot to localize text in quiz result "Correct"		3	Localize the text also.
		Classroom tag is not localized if the user wants to switch app languages, but is in a, let's say, Finnish class.		1	Localize the text also.
3	**H1-3: Minimize users' memory**	When the login token expires, the user is not notified or anything, it just logs out.		3	If the session expired, notify the user with a toast or something else. Or

	load				redirect to the login page and notify there.
		If the user misses and forgets to click "I know" or "don’t Know", the game doesn’t calculate that to the progress, so the user must find the quiz they missed, or else they can't go forward.		3	Users can't go forward without answering the card, or default to "I don't know."
4	**H1-4: Consistency**	Margin top in profile page is lower than other pages, and the width can be checked, not necessarily.		1	Change the margin top to match other pages.
5	**H1-5: Feedback**	When creating a new Quiz, we get no success message or anything.		2	Add toast
		When creating a new classroom, no success message is shown.		2	Add toast
6	**H1-6: Clearly marked exits**	No usability issues found.		0	

7	**H1-7: Shortcuts**	No shortcuts to press spacebar or enter or - and + to navigate in the quiz game		3	Add event listener for enter, forward, backwards, I don't know and I know.
		No way of bookmarking or favoriting quizzes, or recently played. Users must always search for the quiz. Though it is possible to save the link.		2	Add favorite options, or recently played.
8	**H1-8: Precise & constructive error messages**	If the teacher 's application is rejected, there are no messages shown.		4	Inform the user that the application is rejected and the reason for why it is rejected.
9	**H1-9: Prevent errors**	When a learning material is published, there is no way to delete it or edit it. If a teacher makes a mistake there, it's not revertable or editable.		4	Add an edit button, and delete!
		Destructive actions (leaving a classroom, removing a user from classroom, etc.) do not ask for confirmation, but just do the action immediately.		3	On all destructive actions, prompt the user "Are you sure you want to X?"

		If a quiz creator presses the submit button with all required fields in place, the quiz is immediately published as is, whether it is ready or not. It can be edited afterwards from the own quizzes listing.		2	Prompt a "Publish quiz?" before actually submitting the form.
10	**H1-10: Help and documentation**	No About or FAQ section for new users.		4	Add a dedicated About or FAQ page where there is documentation on how to use the app.
		No tooltips for anything, not even for classroom creation or anything.		4	Add tooltips in complex areas, like classroom creation, quiz creation, and learning material creation.
		On the profile page, a user can request promotion to teacher. However, it is never implied on the form what a teacher can do, why would a user want to request promotion?		2	Add a concise description of what a teacher can do more than a regular user can, on top of the request promotion form.

Summary of Findings

After thorough testing, the team managed to find a total of 20 issues. Many of the issues were minimal, but the team also found significant issues that could severely damage the user experience of the application. One of the areas that had the most issues according to the heuristics was the help and documentation section. The team found that the application has no documentation or help. Other severe ones were in the prevention of errors and error messaging. The table below represents the issue totals by severity:

Issue Summary	Count
Total issues	**20**
0 - Not a usability problem	1
1 - Cosmetic problem	4
2 - Minor usability problem	5
3 - Major usability problem	6
4 - Usability catastrophe	4

The team created a bug and issue table, where they track the issues and mark the date they are fixed. With this style of reporting and bookkeeping, the team can make sure that no issues are left unresolved. The table can be found in the github repository under /doc folder.